TAB 13.4 IUPAC NAMES FOR ALKENES

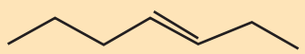
S No	Structure	IUPAC name
1	$CH_3-CH=CH_2$	propene
2	$CH_3-CH_2-CH=CH_2$	but-1-ene
3	$CH_3-CH=CH-CH_3$	but-2-ene
4	$CH_2=C(CH_3)-CH_3$	2-methyl prop-1-ene
5	$CH_2=CH-CH(CH_3)-CH_2-CH_3$ (1 2 3 4 5)	3-methyl pent-1-ene

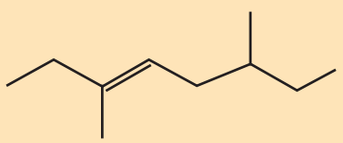
Evaluate Yourself

9. Write the IUPAC names for the following alkenes.

i) $CH_3-CH(CH_3)-CH=CH-CH_3$

ii) $CH_3-C(CH_2-CH_3)=CH-CH_2-CH_3$

iii) 

iv) 

10. Draw the structures for the following alkenes.

i) 6 – Bromo – 2,3 – dimethyl – 2 – hexene

ii) 5 – Bromo – 4 – chloro – 1 – heptene

iii) 2,5 – dimethyl – 4 – octene

iv) 4 – Methyl – 2 pentene

(ii) Isomerism:

Presence of double bond in Alkene provides the possibility of both structural and geometrical isomerism.

Structural Isomerism:

The first two member's ethene C_2H_4 and propene C_3H_6 do not have isomers because the carbon atoms in the molecules can be arranged only one distinct way. However from the third member of alkene family butene C_4H_8, structural isomerism exists.

(i) $CH_3-CH_2-CH=CH_2$ but-1-ene

(ii) $CH_3-CH=CH-CH_3$ but-2-ene

(iii) $CH_2=C(CH_3)-CH_3$ 2-methyl prop-1-ene

structures (i) & (ii) are position isomers. structures (i) & (iii), (ii) & (iii) are chain isomers.

Evaluate Yourself

11) Draw the structure and write down the IUPAC name for the isomerism exhibited by the molecular formulae:

(i) C_5H_{10} – Pentene (3 isomers)

(ii) C_6H_{12} – Hexene (5 isomers)

Geometrical isomerism:

It is a type of stereoisomerism and it is also called cis-trans isomerism. Such type of isomerism results due to the restricted rotation of doubly bounded carbon atoms.

If the similar groups lie on the same side, then the geometrical isomers are called Cis-isomers. When the similar groups lie on

11th Chemistry Chapter-13 Vol-2 EM.indd 191 09-01-2023 14:44:52